Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 05: Python Programming I

Digital for Executive

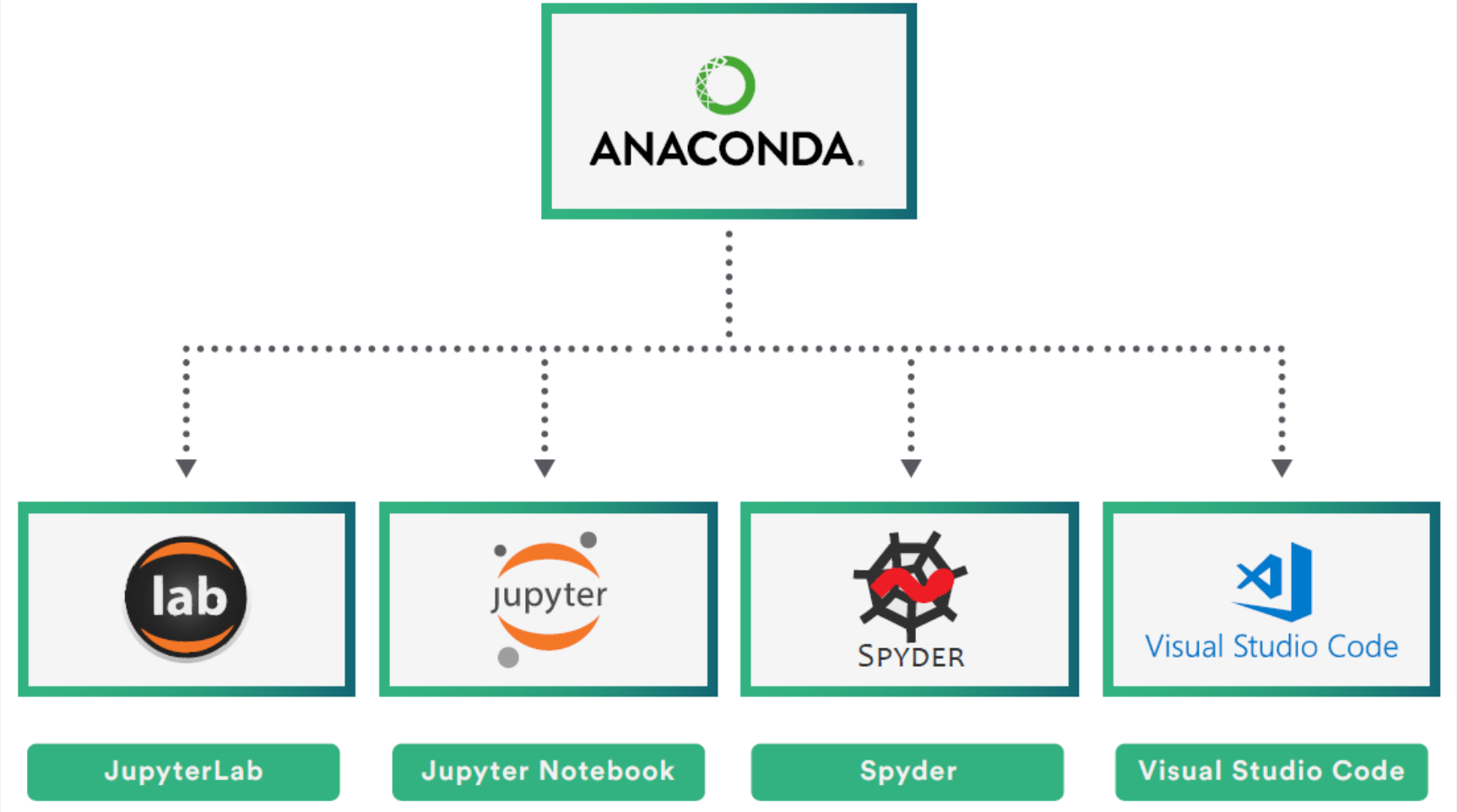
Outline

- Running Code & Python Output
- Introduction to Variable and Data Types
- Python Conditional Statements
- Exercise

Running Code & Python Output

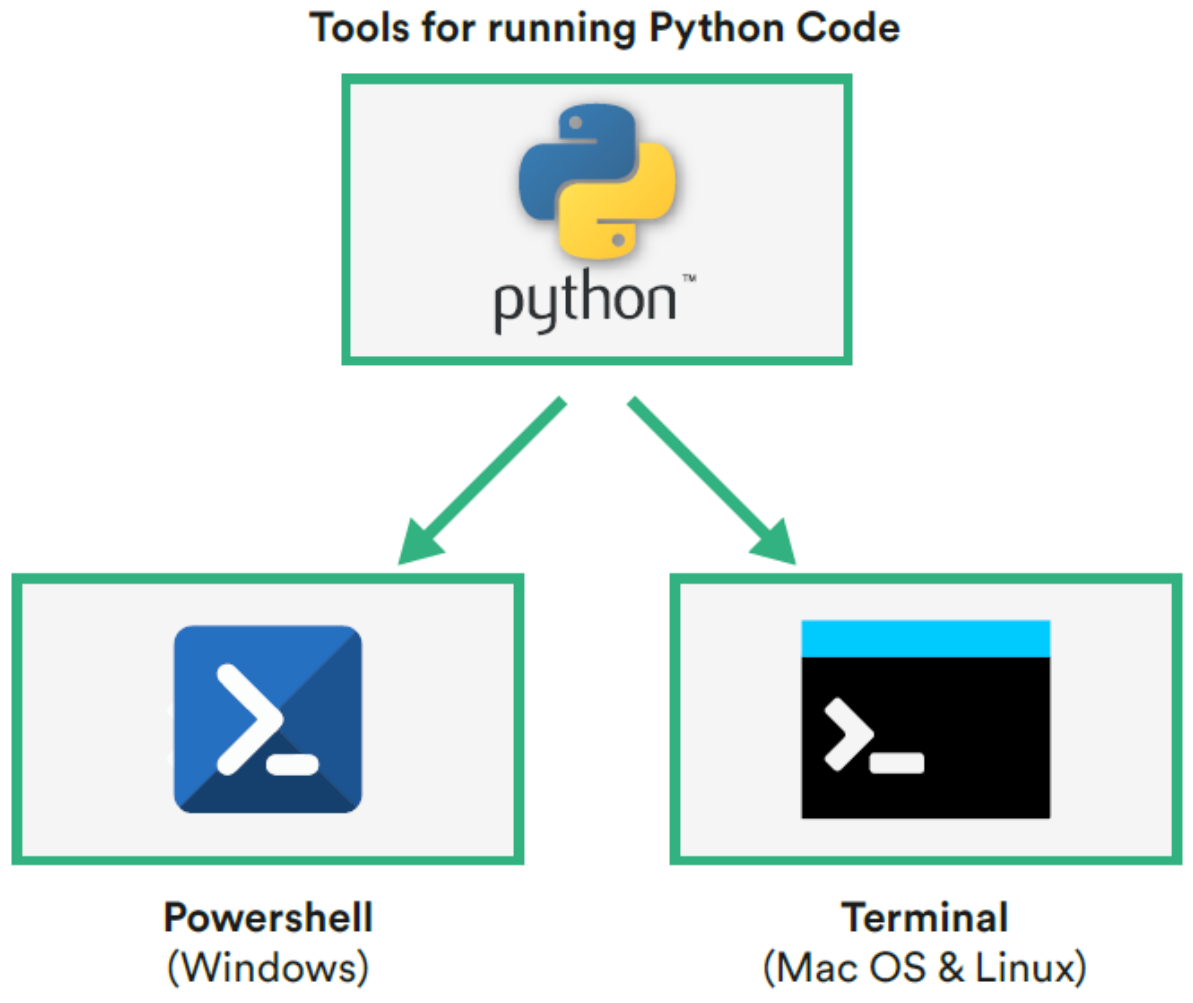
Tools

Tools


ANACONDA.
lab
jupyter
SPYDER
Visual Studio Code
JupyterLab
Jupyter Notebook
Spyder
Visual Studio Code

Terminal / PowerShell

Terminal / PowerShell



Python Output (Print Function)

Our First Code

Python print() function

print() function berfungsi untuk mencetak pesan yang ditentukan ke layar, atau terminal dimana kita menjalankan programnya.

Pesan bisa tipe data string, atau tipe data lainnnya, dimana kalau tipe data lain akan diubah menjadi string sebelum ditulis ke layar.

Notes : function penjelasan simplenya adalah sebuah fungsi/alat yang diprogram untuk tujuan tertentu.

Comment

Comment

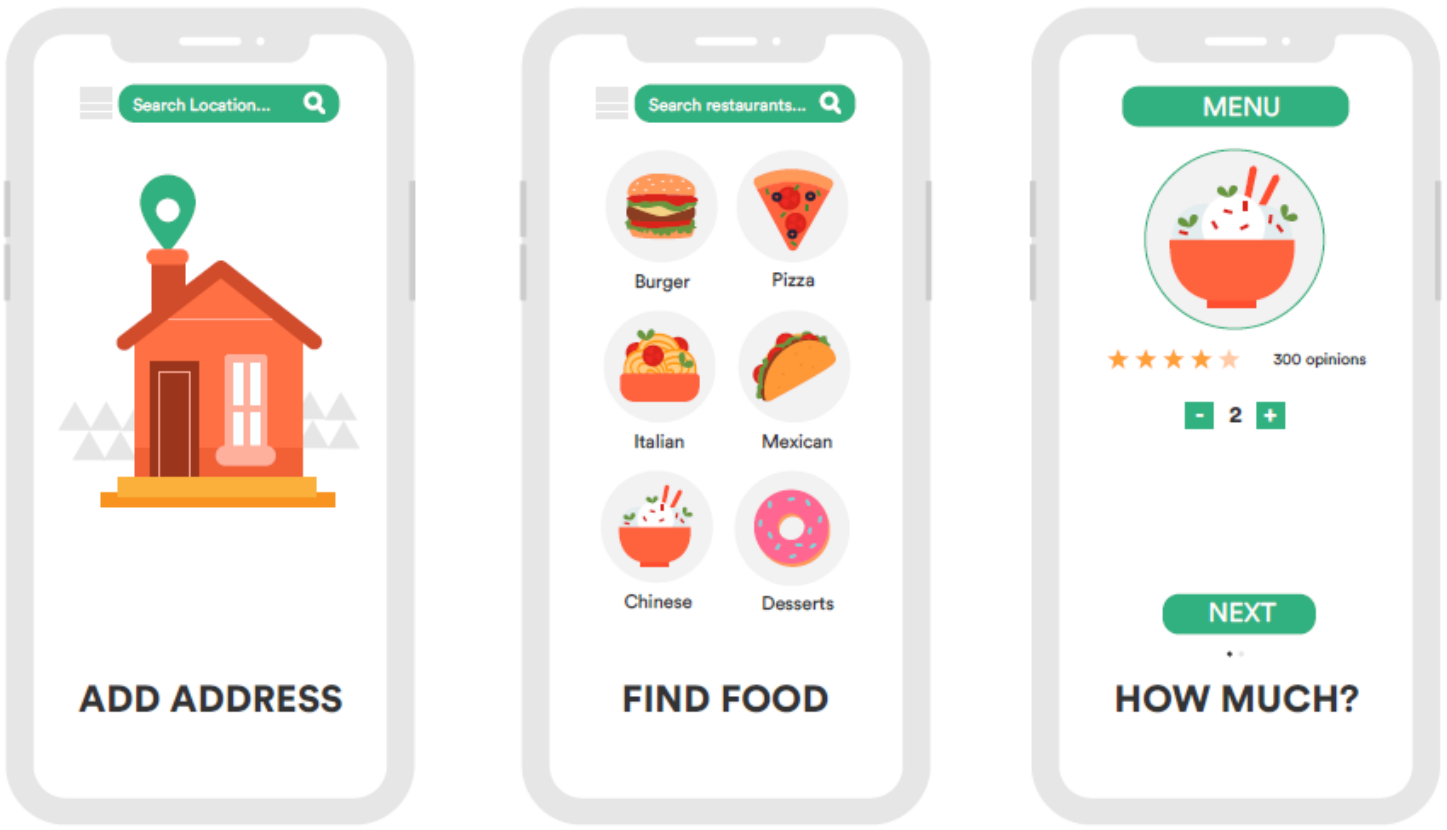
firstcode.py

firstcode.py

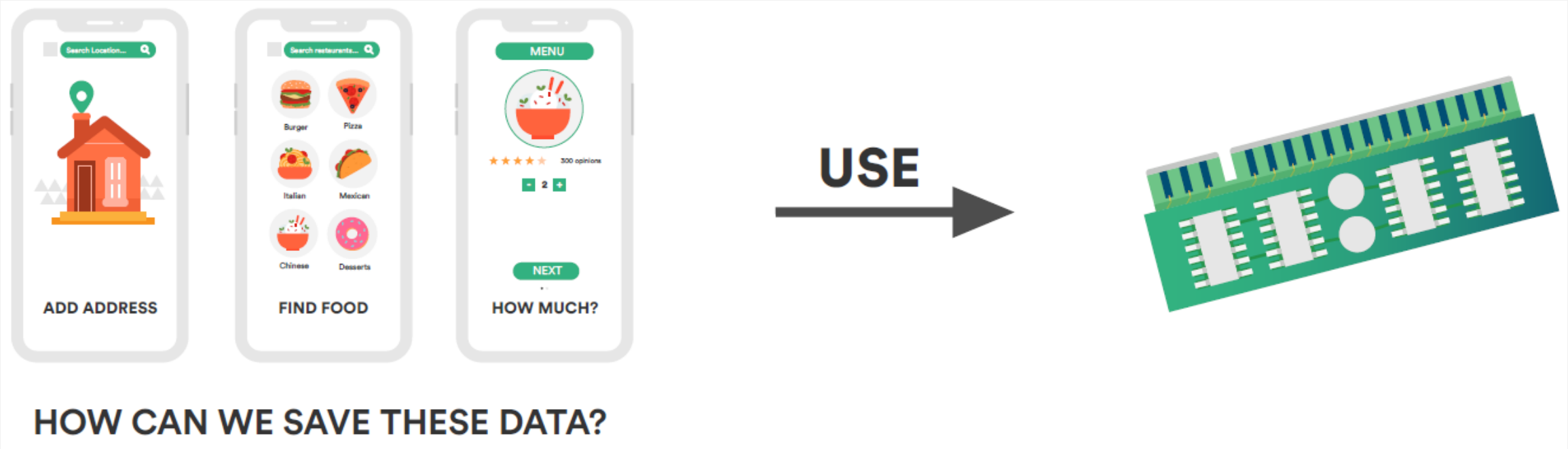
```python
# Dibawah ini contoh penggunaan function print
print('Selamat!') # Text Selamat! akan muncul di terminal
# Semua text yang dimulai dengan tanda pagar
# adalah sebuah comment, dan bukan code python
# jadi tidak akan dijalankan

```

Introduction to Variables and Data Types

Variable Introduction

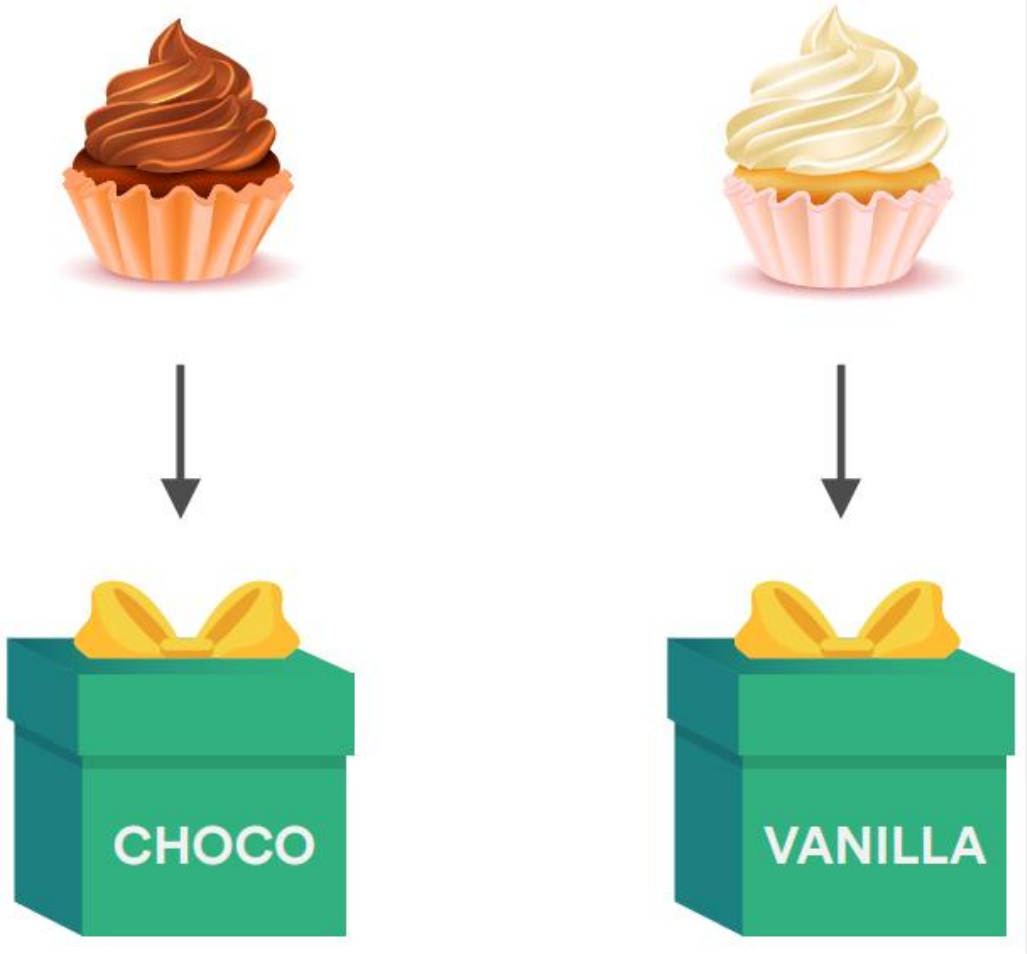


Variable Introduction



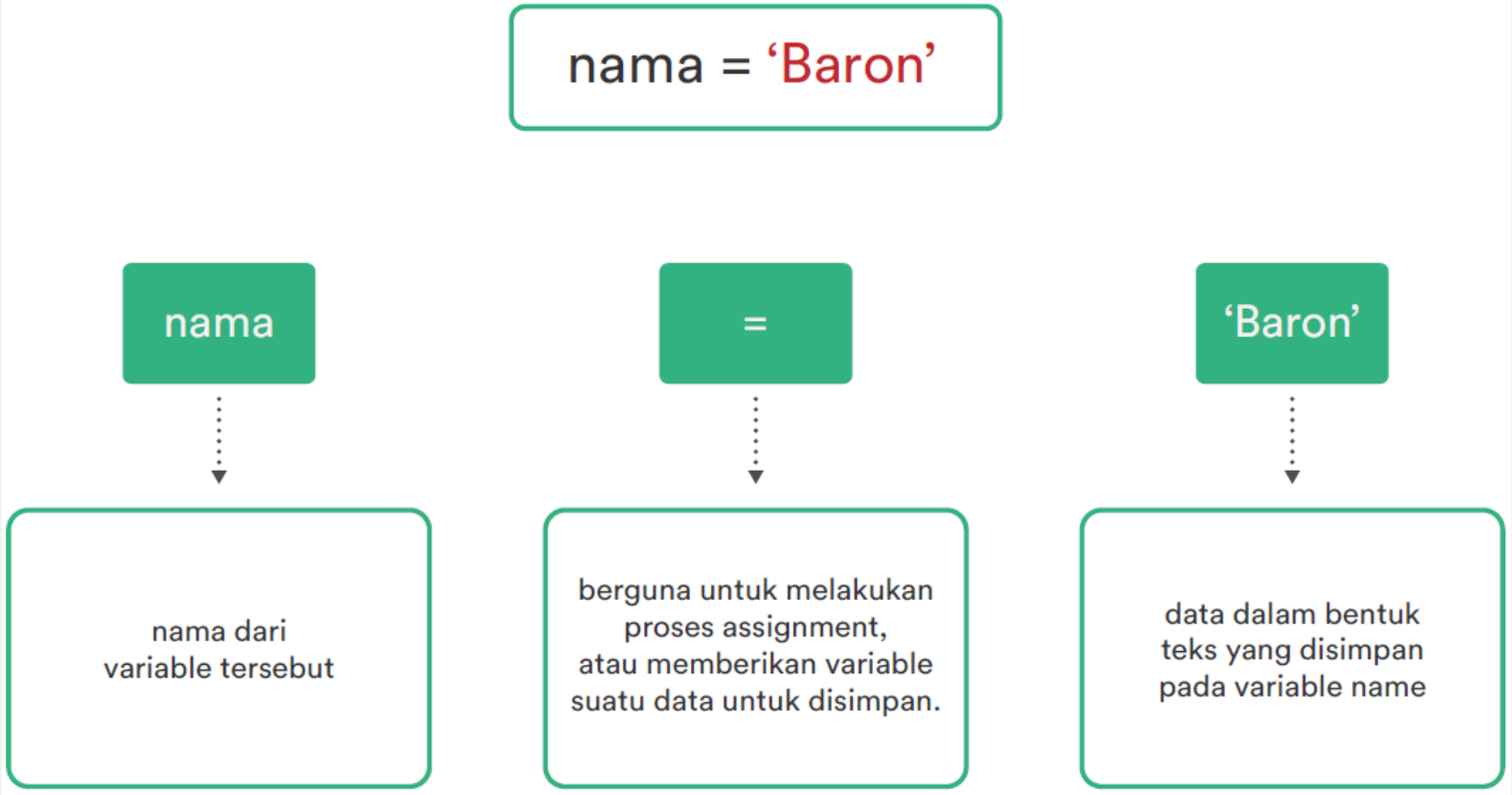
What is Variable?

What is Variable



What is Variable

What is Variable



Rules of Variables

Rules of Variables

Dalam membuat nama dari variable ada beberapa hal yang harus kita perhatikan, pertama, variable bisa memiliki nama yang pendek seperti x dan y atau juga bisa nama yang lebih deskriptif seperti umur, nama Mobil, dan total_income.

Berikut kumpulan aturan dalam penamaan variable di Python:

- Nama variable harus dimulai dengan huruf atau karakter garis bawah (underscore)
- Nama variable tidak bisa dimulai dengan angka
- Nama variable hanya boleh berisi karakter alfa-numerik dan garis bawah (A-z, 0-9, dan _)
- Nama variable itu case-sensitive (nama, Nama dan NAMA adalah 3 variable berbeda)

Rules of Variables

EXAMPLES

```
#Legal variable names:
myvar = "Baron"
my_var = "Baron"
_my_var = "Baron"
myVar = "Baron"
MYVAR = "Baron"
myvar2 = "Baron"
```

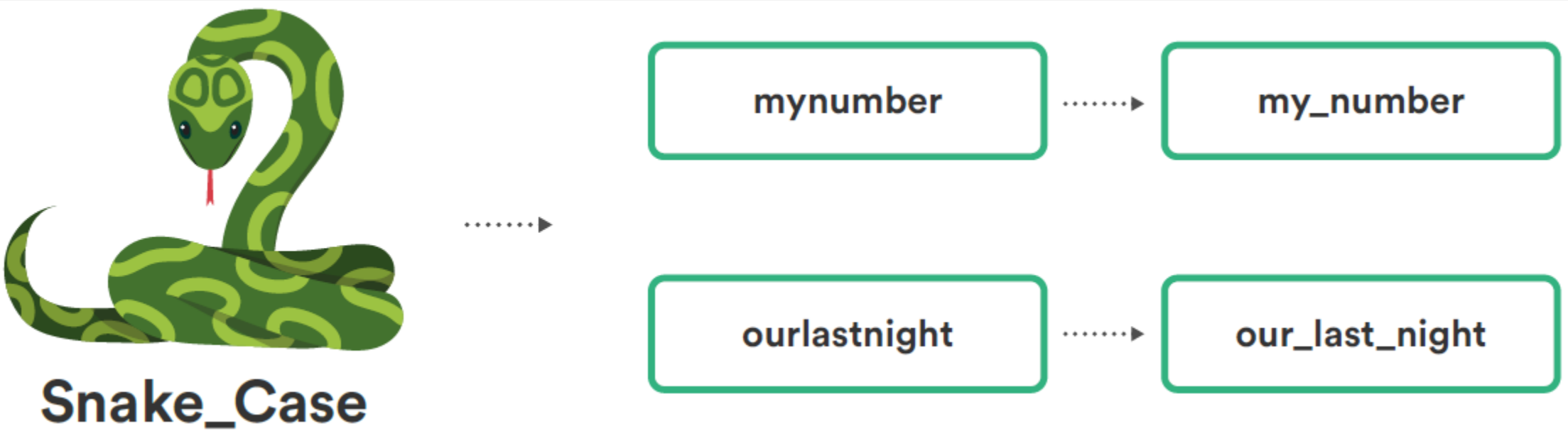
```
#Illegal variable names:
2myvar = "Baron"
my-var = "Baron"
my var = "Baron"
```

Case Style

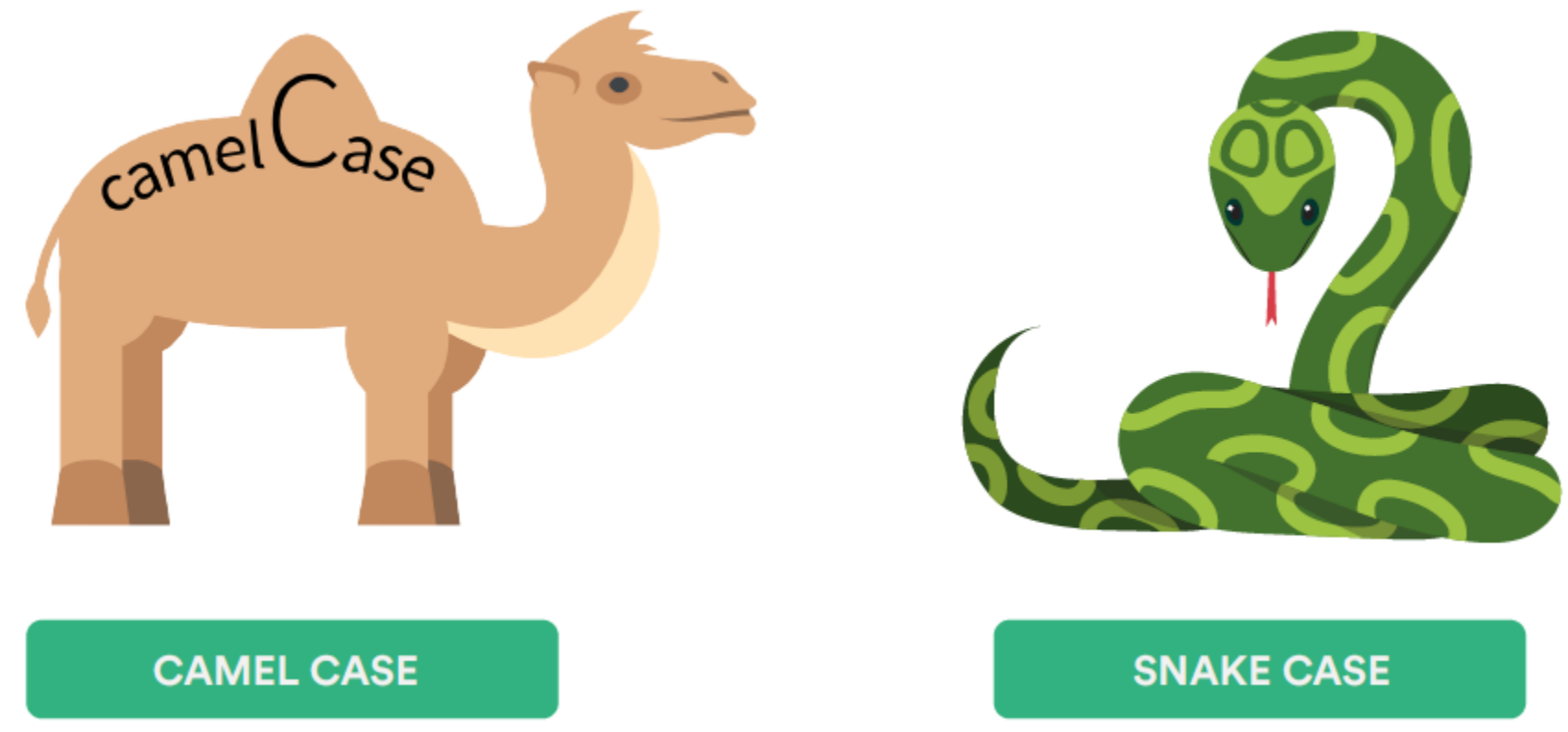
Case Style

Case Style

Case Style

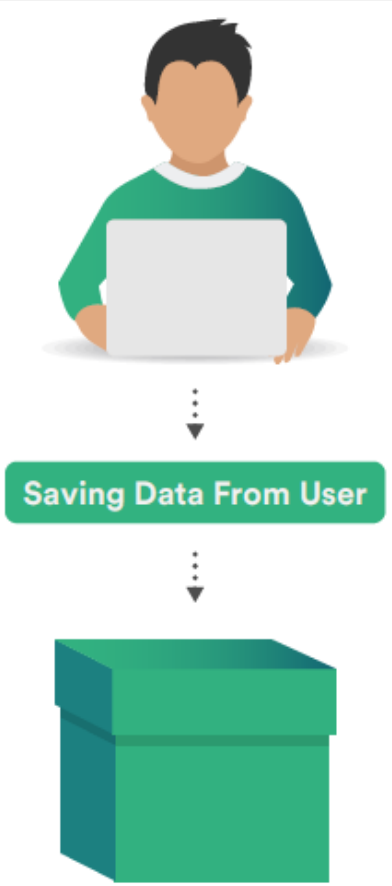


Case Style



Why and When

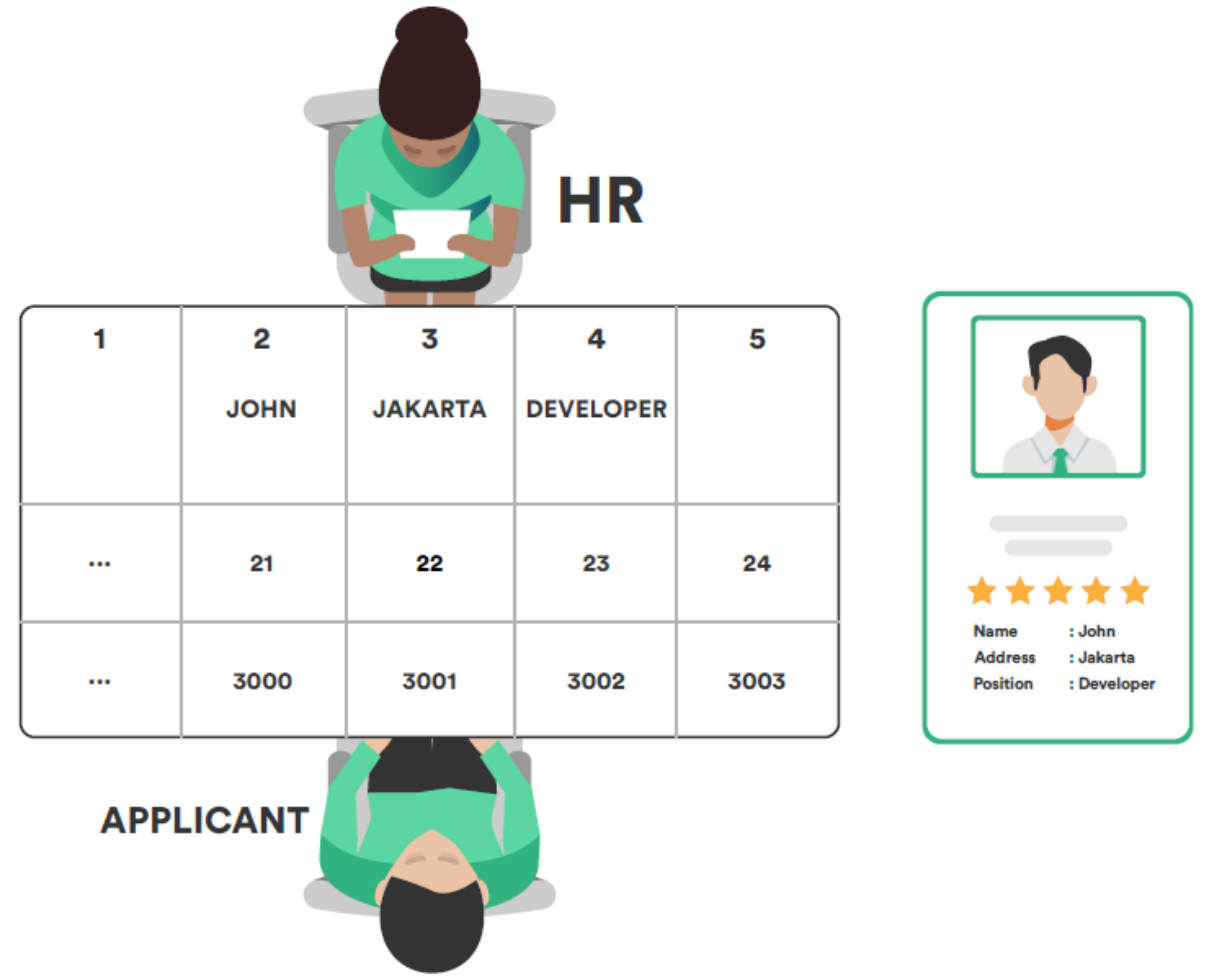
Why and When



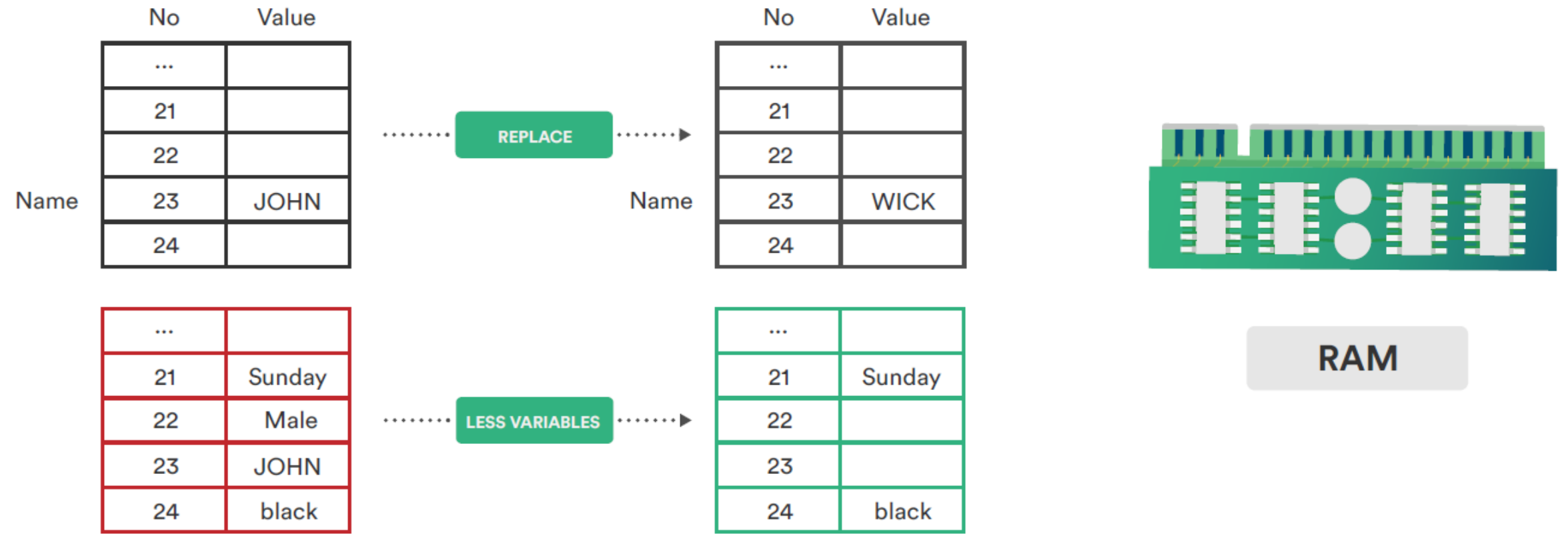

```
# pengulangan
print('Terima kasih sudah datang')
print('Terima kasih sudah datang')
print('Terima kasih sudah datang')

# dengan variable
pesan = 'Terima kasih sudah datang'
print(pesan)
print(pesan)
print(pesan)
```

How RAM Stores Data

How RAM Stores Data



How RAM Stores Data



Data Types

Python Data Types

Dalam pemrograman, tipe data merupakan hal yang penting.
Variabel dapat menyimpan data dengan berbagai tipe, dan tipe yang berbeda dapat melakukan hal yang berbeda dan digunakan untuk tujuan yang berbeda juga.

Tipe data yang biasa digunakan di Python adalah sebagai berikut :

Tipe Text:	str (string)
Tipe Angka:	int (integer), float
Tipe Koleksi:	list, tuple, dict (dictionary)
Tipe Range:	range
Tipe Set:	set
Tipe Boolean:	bool (boolean)
Tipe None:	None

```
x = "Hello Purwadhika"                #str (string)
x = 'jeruk'                           #str (string)
x = "1234"                            #str (string)
x = 23                                #int (integer)
x = 23.5                              #float
x = ['jeruk', 'apel', 'mangga']       #list
x = ('jeruk', 'apel', 'mangga')       #tuple
x = {"nama" : "Baron", "umur" : 23}   #dict (dictionary)
x = range(7)                          #range
x = {"jeruk", "apel", "mangga"}       #set
x = True                              #bool (boolean)
x = False                             #bool (boolean)
x = None                              #None
```

Python Data Types

Tipe Text: str (string)

```
x = "Hello Purwadhika"      #str (string)
x = 'jeruk'                 #str (string)
x = "1234"                  #str (string)
```

Python Data Types

Tipe Angka: int (integer),float

```
x = 23          #int (integer)
x = 23.5        #float
```

Python Data Types

Tipe Boolean: bool(boolean)

```
x = True        #bool (boolean)
x = False       #bool (boolean)
```

Python Data Types

Tipe Koleksi: list, tuple, dict(dictionary)

```
x = ['jeruk', 'apel', 'mangga']          #list
x = ('jeruk', 'apel', 'mangga')          #tuple
x = {"nama" : "Baron", "umur" : 23} #dict (dictionary)
```

Python Data Types

Tipe Range: range

```
x = range(7)                    #range
```

Python Data Types

Tipe Set: set

```
x = {"jeruk", "apel", "mangga"}      #set
```

Python Data Types

Tipe None:	None
x = None	#None

Python Data Types

```
x = "Aku string"
xDataType = type(x)
print(xDataType)
```

Numbers, Arithmetic & Assignment Operators, and math Module

Numbers

```
# integer
a = 3
# float
b = 3.14

# 123 x 10^6
c = 123e6
# 123 x 10^-6
d = 123e-6
```

Python Arithmetic Operators

Operator	Name	Example
+	Addition	x + y
-	Subtraction	x - y
*	Multiplication	x * y
/	Division	x / y
%	Modulus	x % y
**	Exponentiation	x ** y
//	Floor division	x // y

Modulus

10 % 2 = 11 11 11 11 11 = 0 (tidak memiliki sisa)

10 % 3 = 111 111 111 1 = 1 (sisa 1)

10 % 4 = 1111 1111 11 = 2 (sisa 2)

Python Assignment Operators

Operator	Example	Same As
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
%=	x %= 3	x = x % 3
//=	x //=3	x = x // 3
**=	x **= 3	x = x ** 3

math Module

Python memiliki modul bawaan yang dapat digunakan untuk kebutuhan operasi matematika.
Modul math ini memiliki sekumpulan methods dan constants.
Info :

- Method adalah sebuah function atau alat (seperti function print dan type yang sudah kita gunakan) yang terdapat pada suatu object
- Object adalah suatu kumpulan data (variable) dan method (function) yang bekerja pada data tersebut
- Constant adalah sebuah variable yang menyimpan suatu nilai yang tidak boleh berubah

math Methods & Constants

Methods

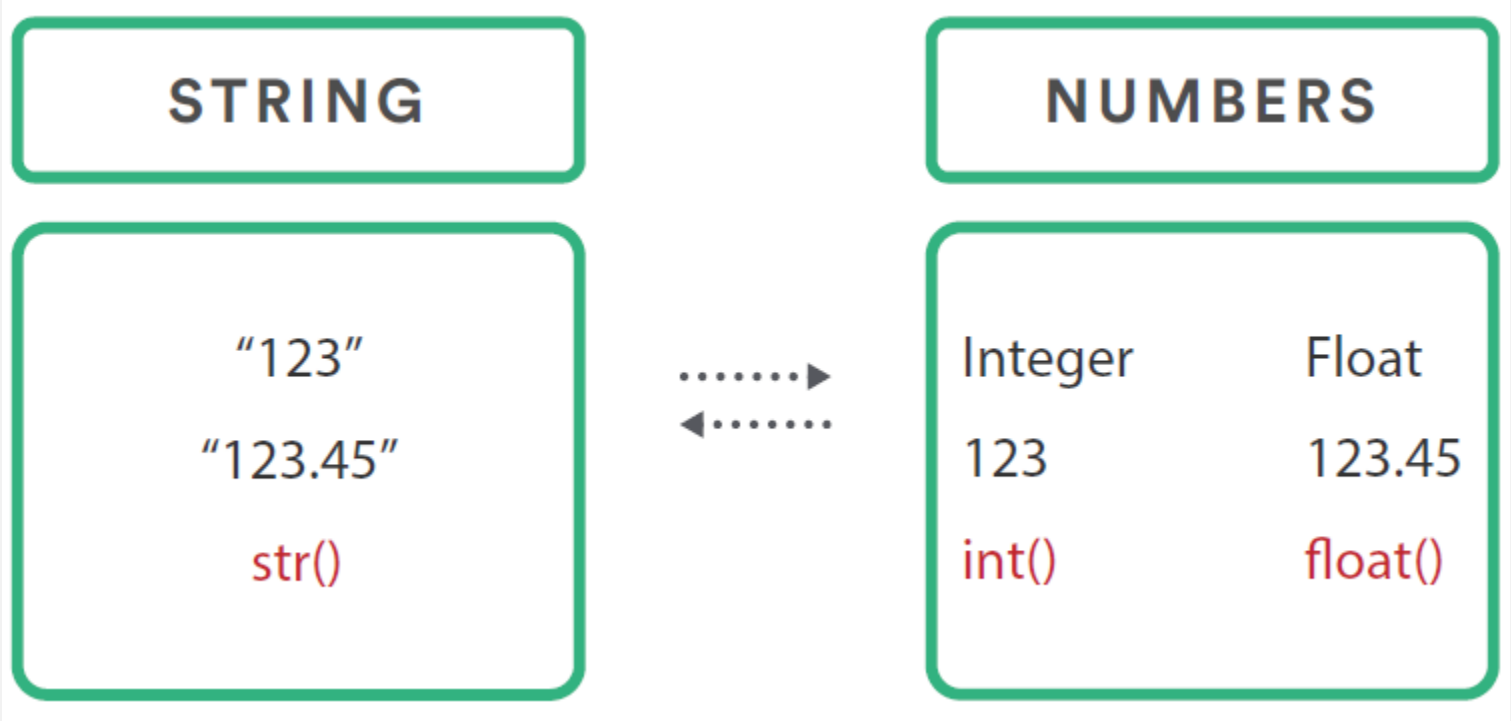
math.ceil(x)	Rounds a number up to the nearest integer
math.fabs(x)	Returns the absolute value of a number
math.floor(x)	Rounds a number down to the nearest integer
math.pow(x, y)	Returns the value of x to the power of y
math.sqrt(x)	Returns the square root of x

Constants

math.pi	Returns PI (3.1415...)

Casting

Python Casting



String

String

"Happy Weekend !"

"23 adalah bilangan ganjil"

"1234" ····· ▶ #Ini bukan int/integer

"True" ····· ▶ #Ini bukan bool/boolean

String Quotes

```
kutipSatu = 'Kutip Satu ("Im Batman")'
kutipDua = "Kutip Dua (I'm Iron Man)"
kutipTiga = '''
        "Im Batman"
        I'm Iron Man
'''
print (kutipSatu)
print (kutipDua)
print (kutipTiga)
```

Escape Character

Untuk menyisipkan karakter yang ilegal dalam sebuah string, gunakan escape character. Escape character adalah garis miring terbalik \ diikuti dengan karakter yang ingin kita sisipkan.

Contoh karakter ilegal adalah tanda kutip dua di dalam string yang diapit tanda kutip dua:

```
# error
txt = "Salam kenal murid "Purwadhika", mari kita belajar bersama."
# escape character jadi tidak error
txt = "Salam kenal murid\"Purwadhika\", mari kita belajar bersama."
```

Escape Character

Code	Result
\’	Single Quote
\\	Backslash
\n	New Line
\t	Tab
\b	Backspace

len Function & String Methods

```
x = 'Halo Dunia'

panjangString = len (x)                    # 10
indexDunia = x.index('Dunia')              # 5
hasilSplit = x.split('  ')                 # ['Halo', 'Dunia']
jadiHurufKecil = x.lower( )                # halo dunia
jadiHurufBesar = x.upper( )                # HALO DUNIA
hurufPertamaCapital = x.capitalize( )      # Halo dunia
```

String Slicing

```
text = "I'm Baron, nice to meet you"
# text[index awal : index akhir]
print (text[1])          # '
print (text[2:])         # m Baron, nice to meet you
print (text[:4])         # I'm
print (text[2:5])        # m B
print (text[:])          # I'm Baron,  nice to meet you
print (text[-1])         # u
print (text[-3:-1])      # yo
```

String Concatenation

Untuk menggabungkan dua string kita dapat menggunakan operator +.

```
a = "Apel"
b = "Jeruk"

c = a + b
print ( c )   # ApelJeruk
d = a + " " + b + "  angga"
print ( d )   # Apel Jeruk Mangga
e = a + " " + str (5)
print ( e )   # Apel 5
```

String Format

Kita tidak bisa menggabungkan string dan angka seperti ini :

```
umur = 23
txt = "nama saya Baron, saya berumur" + umur
print (text)
```

Tapi kita bisa menggabungkan string dan angka dengan menggunakan format() method pada string!
format() method mengambil argumen yang diteruskan, memformatnya,
dan menempatkannya dalam string di mana placeholder {} berada:

```
umur = 23
txt = "nama saya Baron, saya berumur { }"
txt = txt.format(umur)
print (text)
```

String Format

format() method menggunakan jumlah argumen yang tidak terbatas, dan ditempatkan di placeholder masing-masing :

```
jumlahApel = 3
jumlahJeruk = 5
pembelian = "Saya mau beli { } Apel, { } Mangga, dan { } Jeruk."
print (pembelian.format(jumlahApel,7,jumlahJeruk))
```

String Format

Kita bisa menggunakan nama indeks pada placeholder {}
untuk memastikan argumen ditempatkan di placeholder yang benar:

```
jumlahApel = 3
jumlahJeruk = 5
pembelian = "Saya mau beli {jmlApel} Apel, {jmlMangga} Mangga, dan {jmlJeruk} Jeruk."
print (pembelian.format(jmlMangga=7, jmlJeruk=jumlahJeruk, jmlApel=jumlahApel))
```

Check String

Untuk memeriksa apakah frase atau karakter tertentu ada dalam sebuah string, kita dapat menggunakan kata kunci in atau not in.

```
txt = "Nama saya Baron"
x = "aya" in txt
print (x)  #True

txt = "Nama saya Baron"
x = "say" not in txt
print (x)  #False
```

User Input

Python User Input

- Python memiliki kemampuan untuk meminta input dari user.
- Caranya sedikit berbeda di Python 3.6 dengan Python 2.7.
- Python 2.7 menggunakan function raw_input()
- Python 3.6 menggunakan function input()

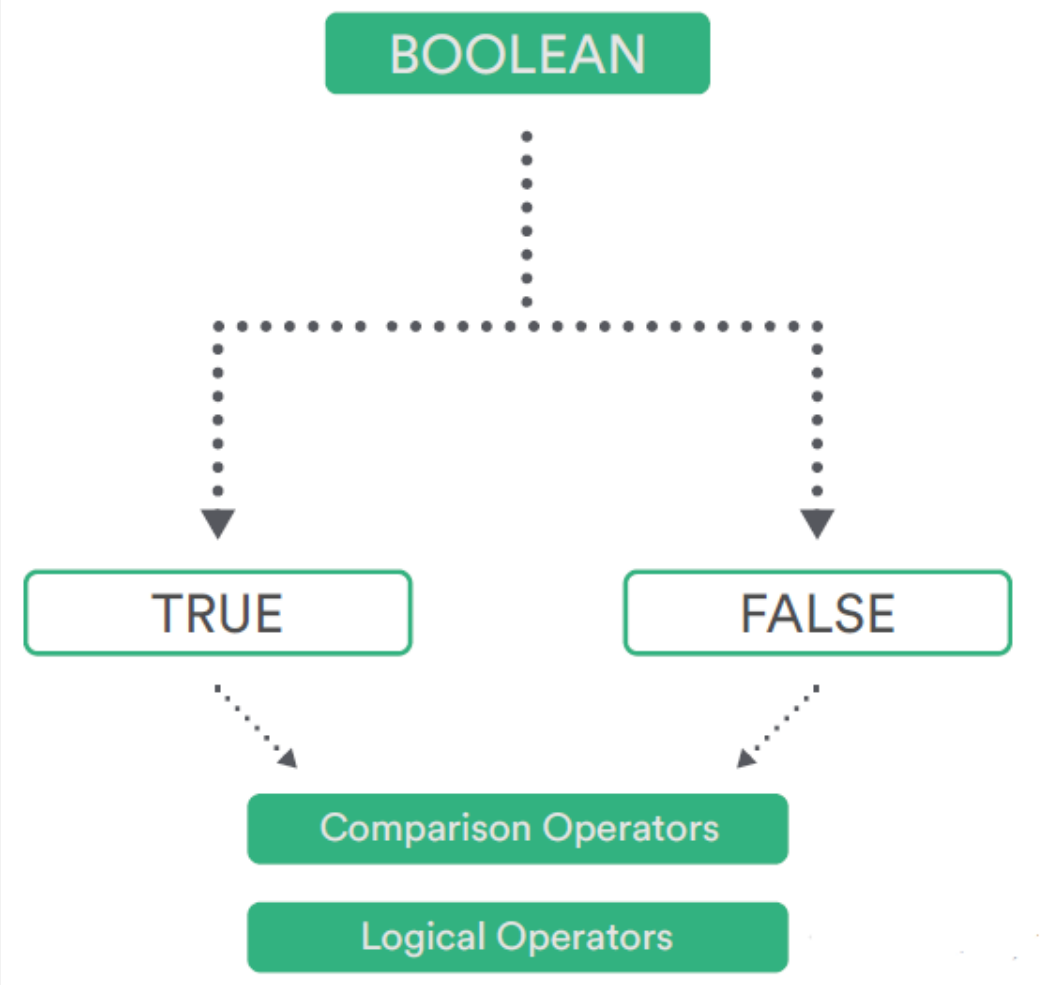
Contoh berikut menanyakan nama pengguna, dan ketika Anda memasukkan nama pengguna, nama pengguna yang anda masukkan akan dicetak di layar:

```
username = input ("Masukkan username:")
print ("Username anda : " + username)
```

Python Conditional Statements

Boolean, Comparison & Logical Operators

Boolean, Comparison & Logical Operators



Boolean, Comparison & Logical Operators

Comparison Operators

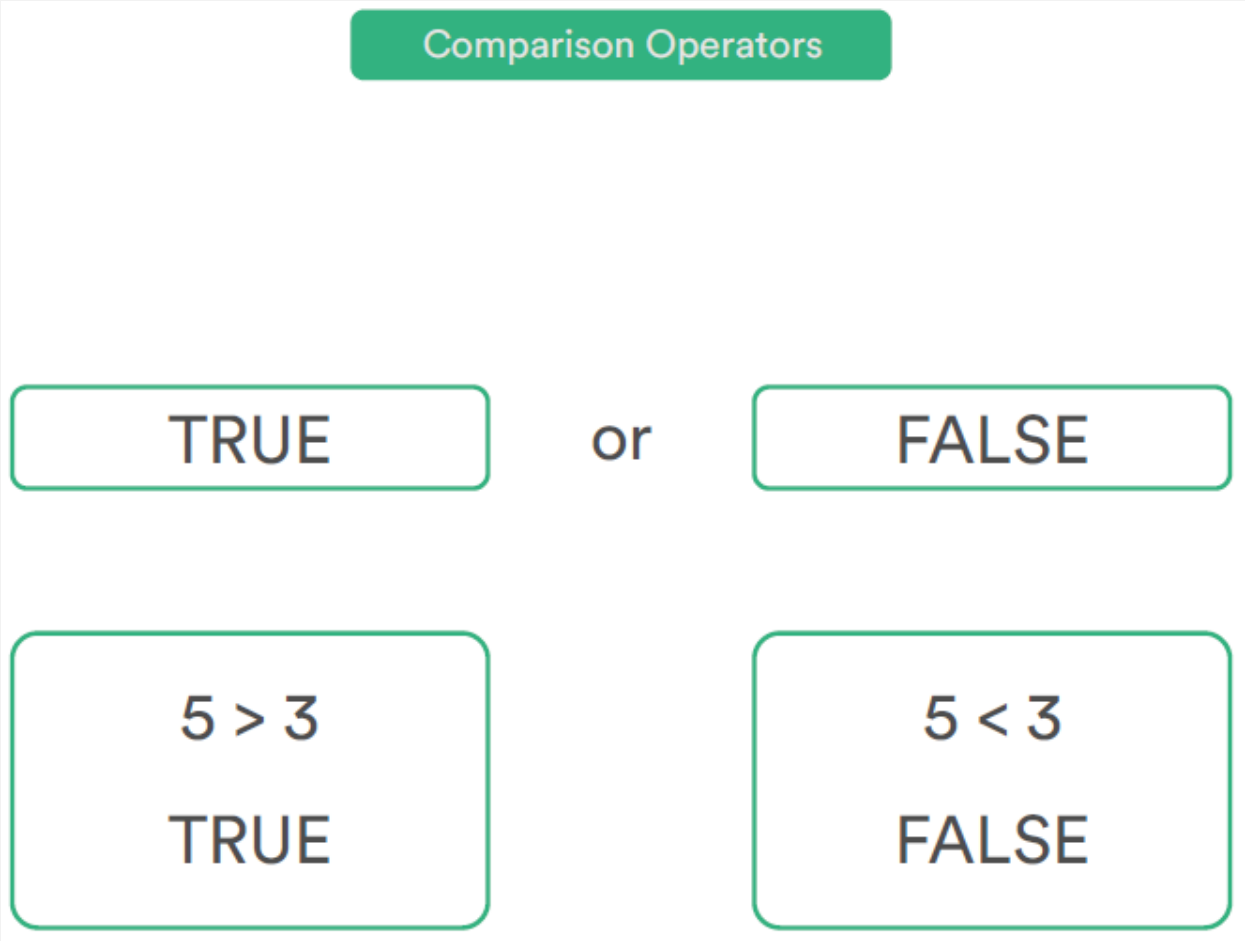
>	greater than
<	less than
>=	greater than or equal to
<=	less than or equal to
!=	not equals
==	equals

Boolean, Comparison & Logical Operators

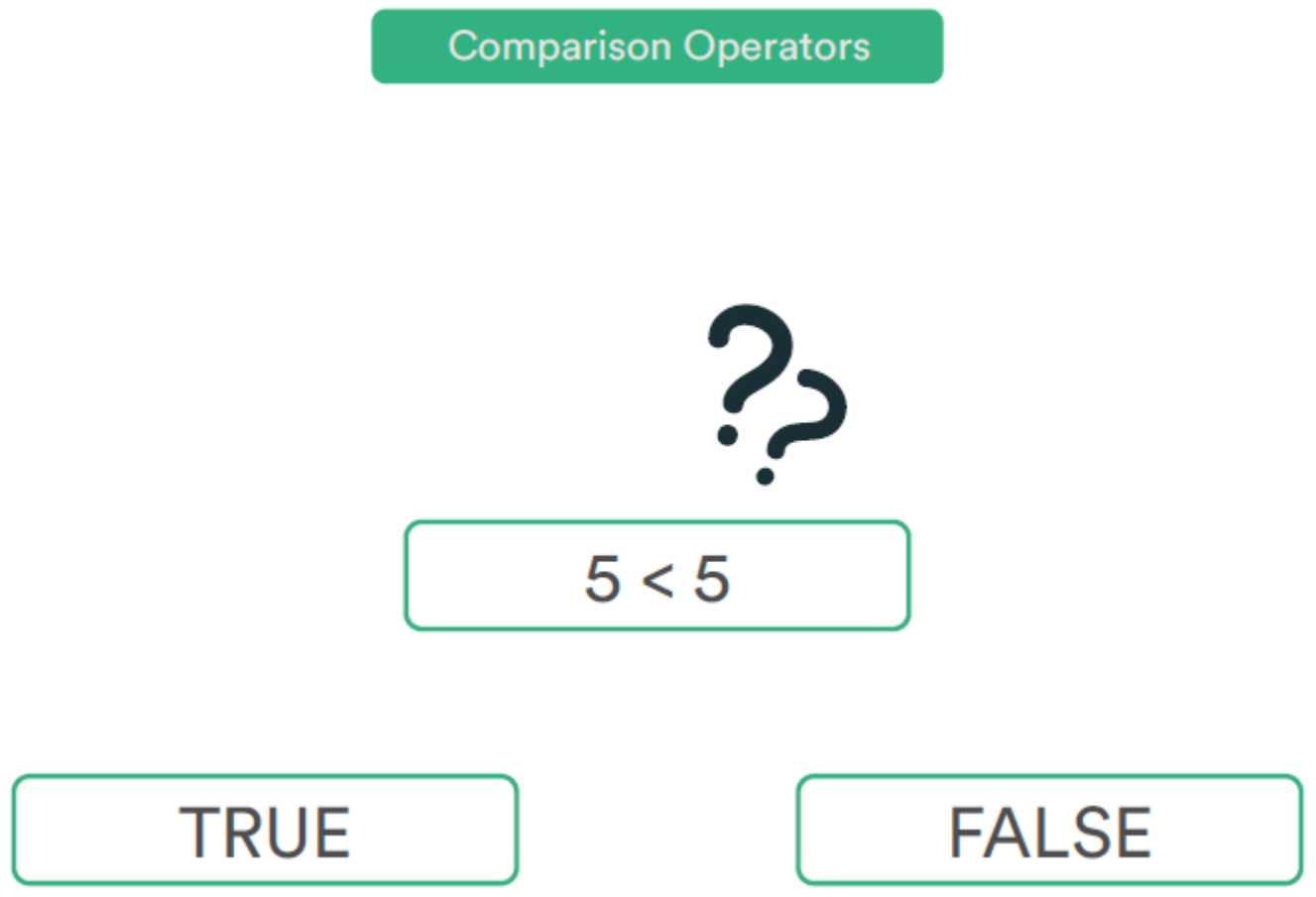
Logical Operators

and	(if both expression are true , result will be TRUE)
or	(if one of the expression true, result will be TRUE)
not	(reverse boolean value TRUE/FALSE)

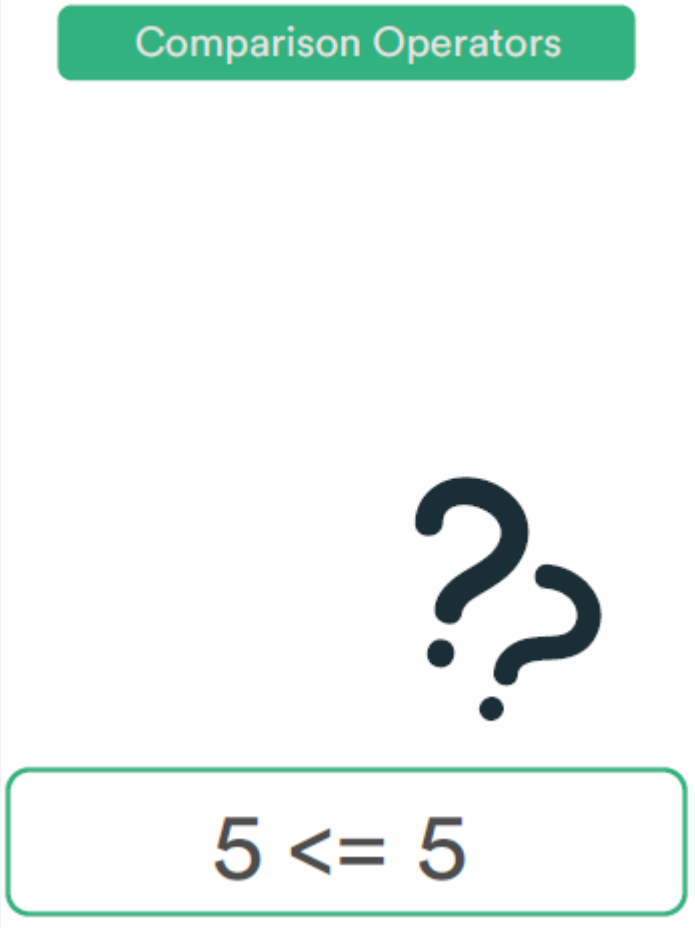
Boolean, Comparison & Logical Operators



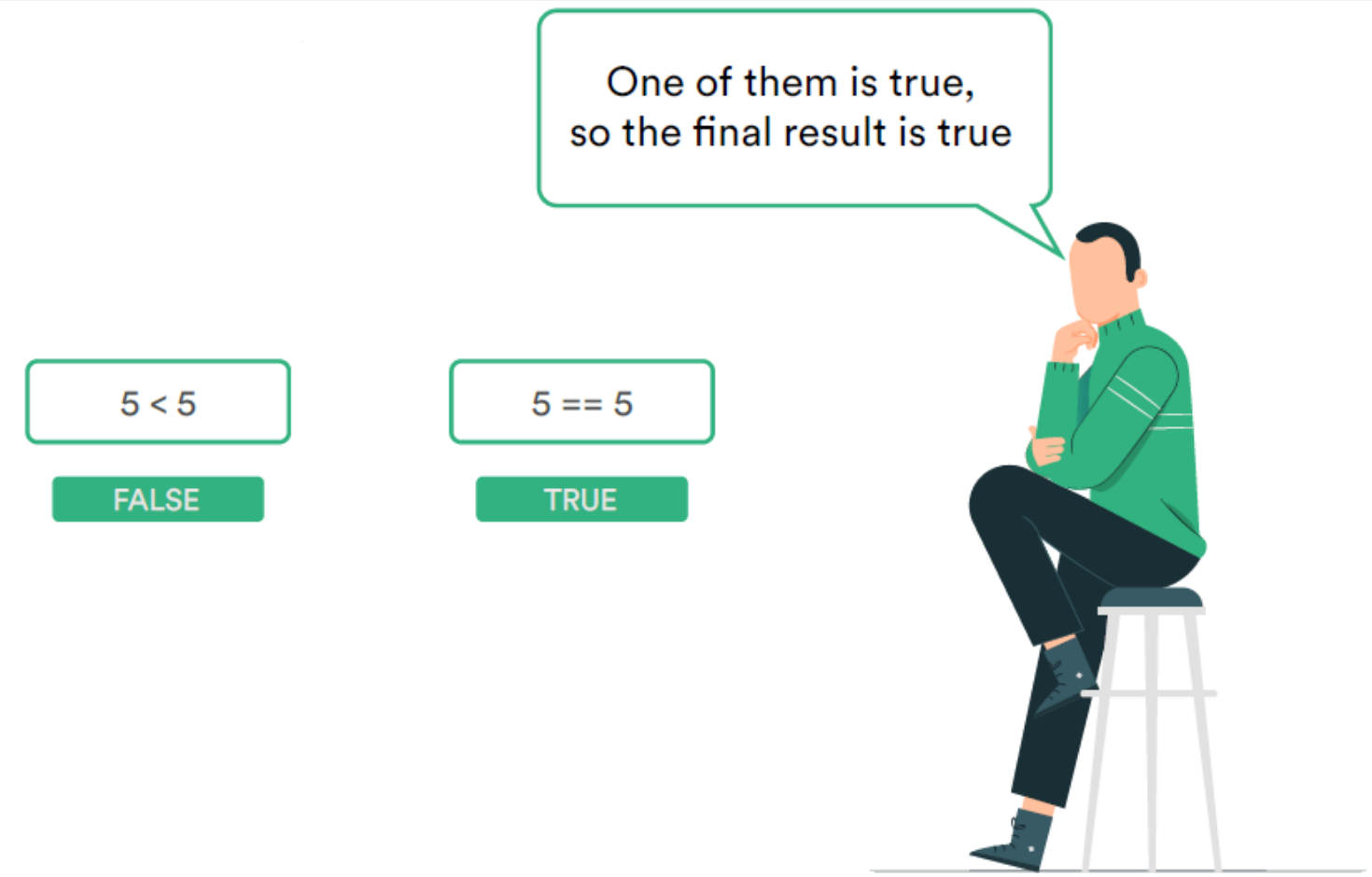
Boolean, Comparison & Logical Operators



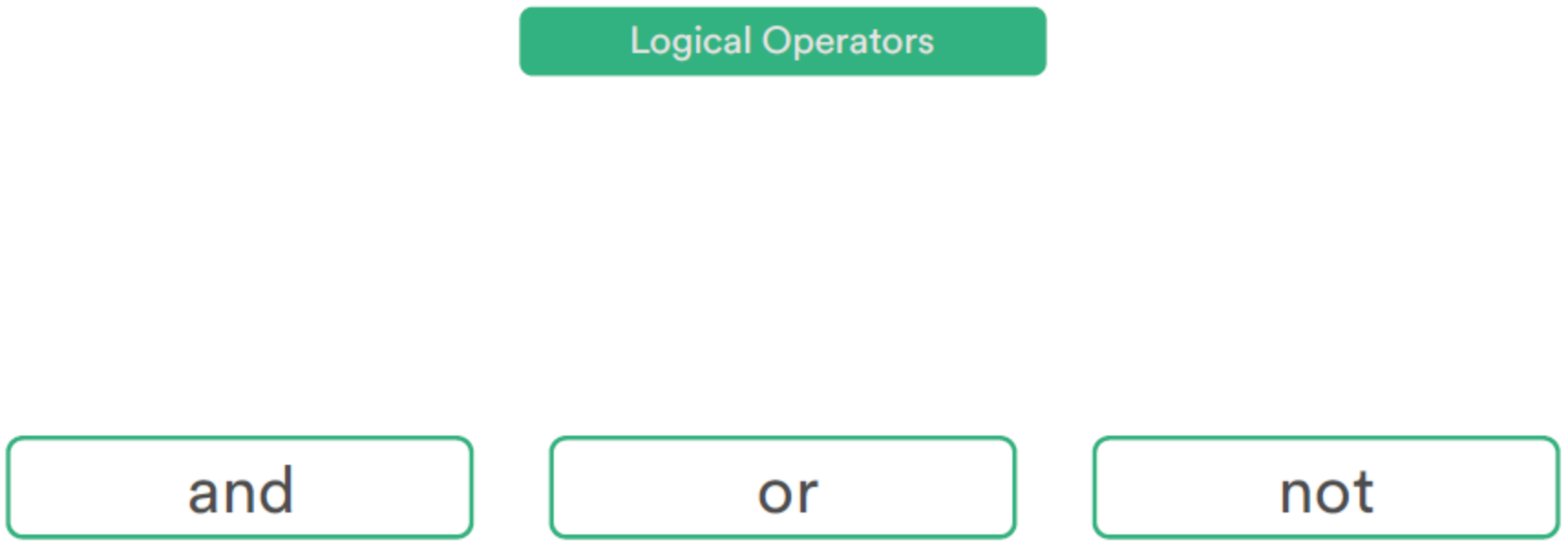
Boolean, Comparison & Logical Operators



Boolean, Comparison & Logical Operators



Boolean, Comparison & Logical Operators



Boolean, Comparison & Logical Operators

Logical Operators

EXAMPLE

True and False	True or False	not(True)	not(False)
False	True	False	True

Boolean, Comparison & Logical Operators

P	Q	P and Q
T	T	T
T	F	F
F	T	F
F	F	F

Boolean, Comparison & Logical Operators

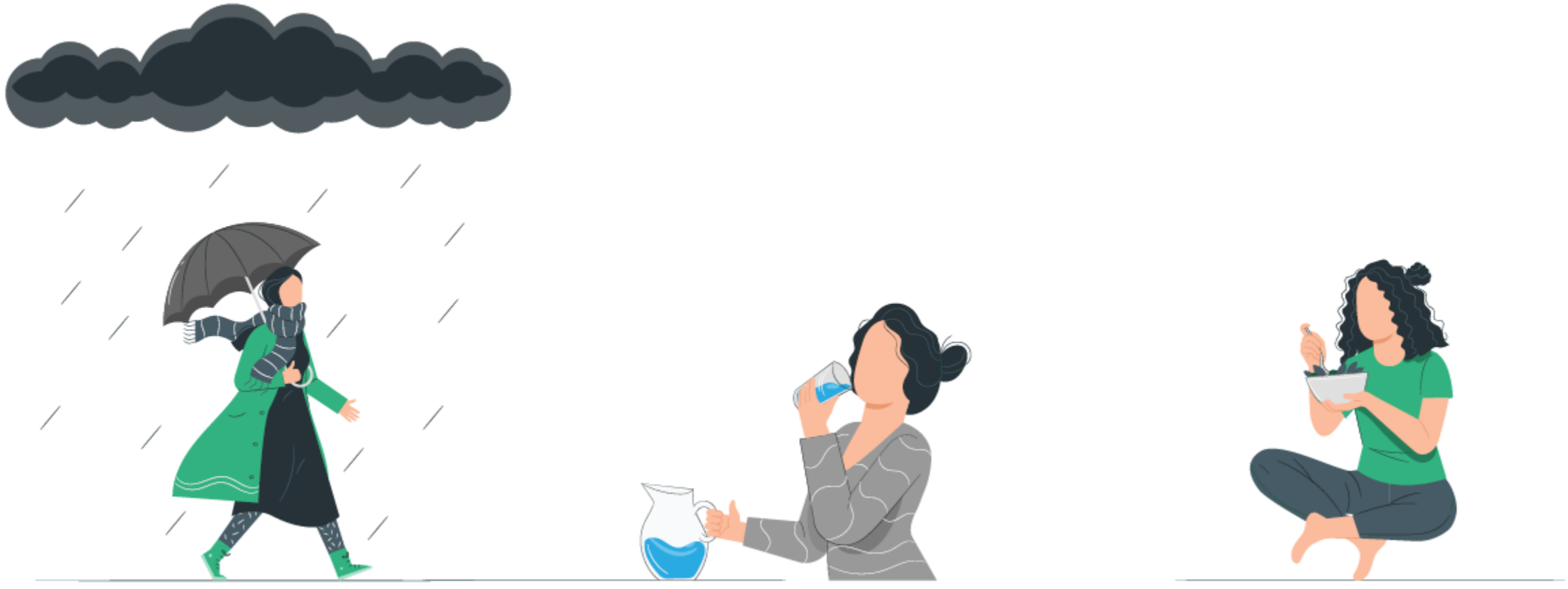
P	Q	P or Q
T	T	T
T	F	T
F	T	T
F	F	F

Boolean, Comparison & Logical Operators

P	not P
T	F
F	T

Conditional Statements Explanation

Conditional Statements Explanation



Conditional Statements Explanation

Conditional Statements Explanation

Conditional Statements Explanation

PATTERN 1

```
if(condition) :
    # do something
```

PATTERN 2

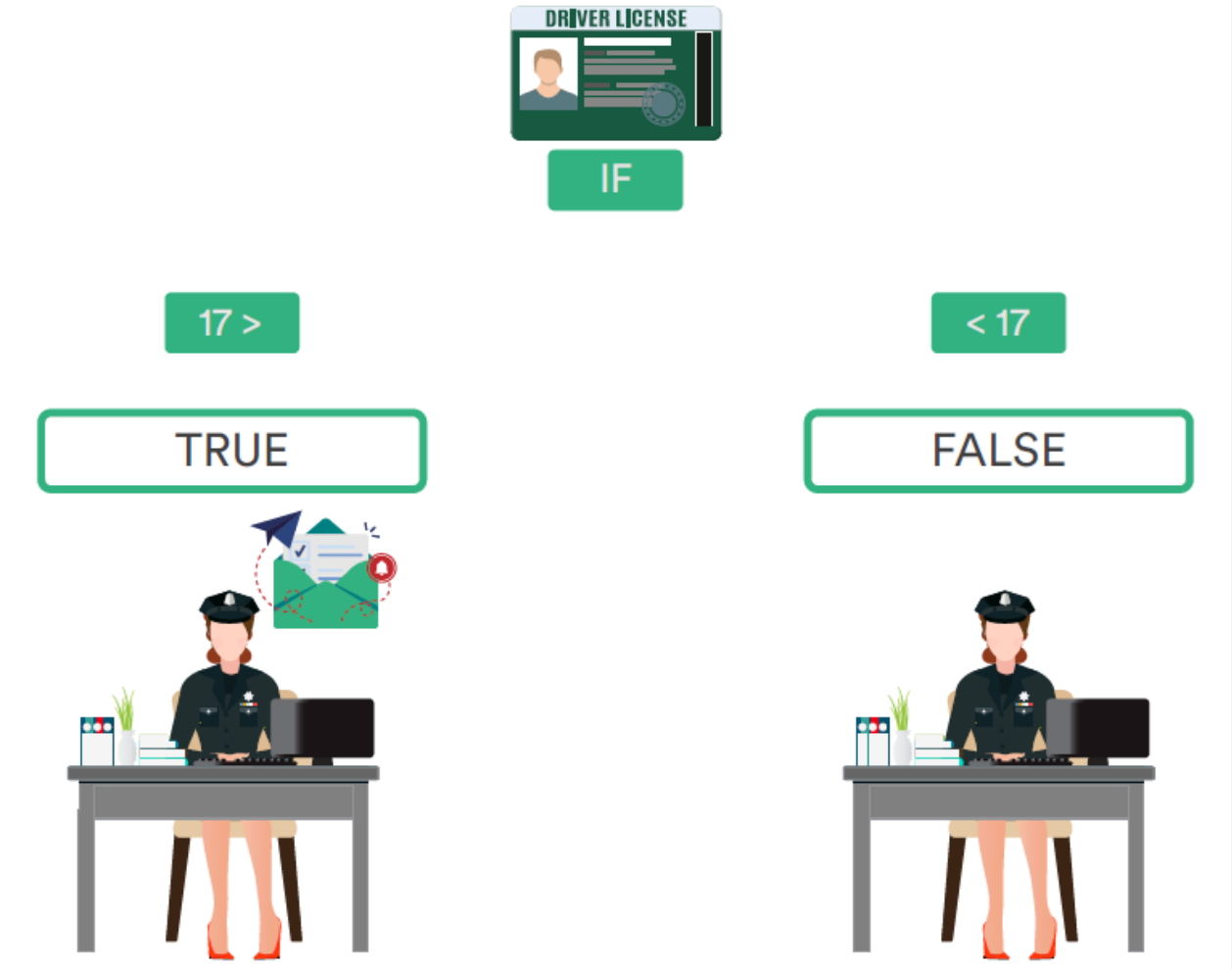
```
if(condition) :
    # do something
else :
    # do other thing
```

PATTERN 3

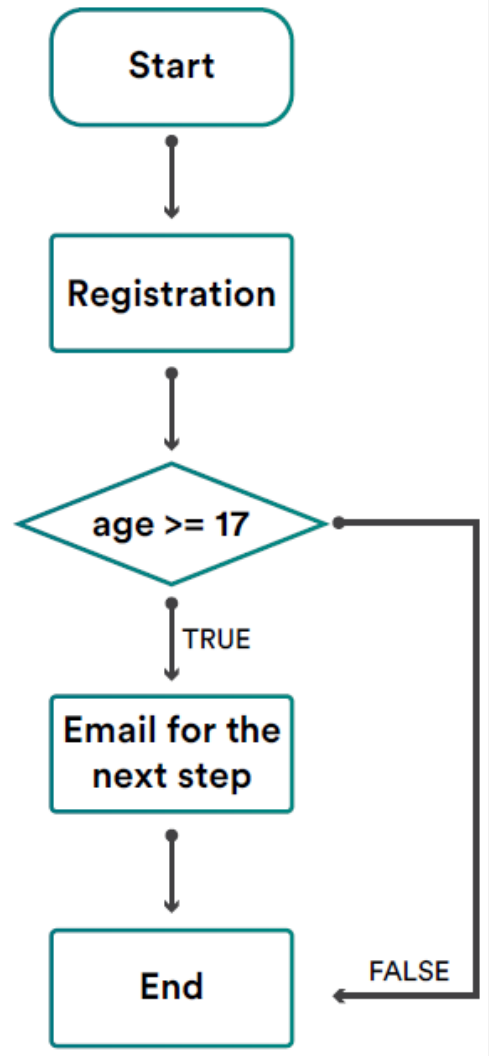
```
if(condition) :
    # do something
elif(condition) :
    # do some other thing
else :
    # do other thing
```

if Statement

if Statement



if Statement



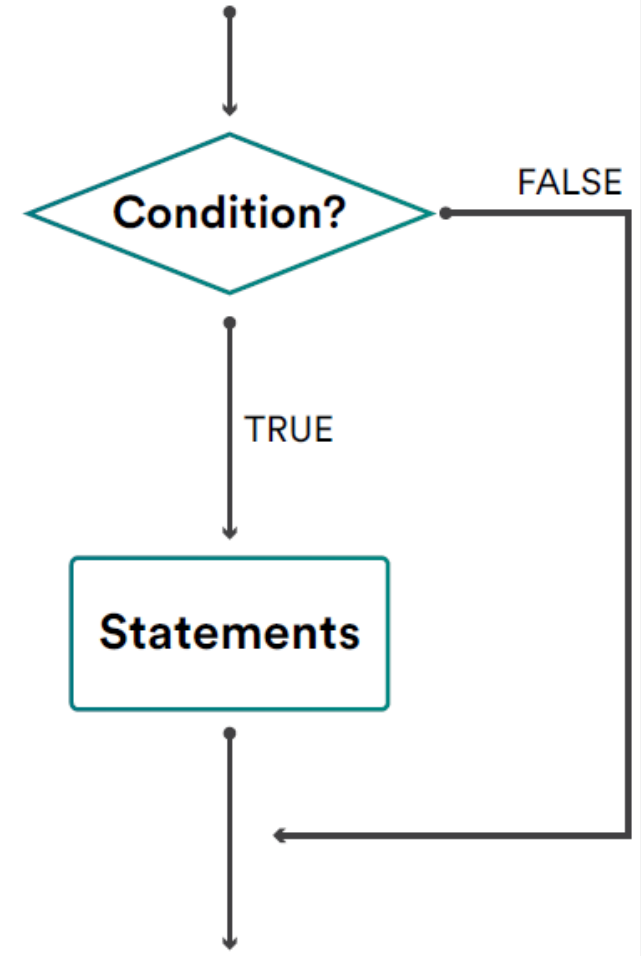
if Statement

SYNTAX

```
if condition :
    statement 1
    statement 2
```

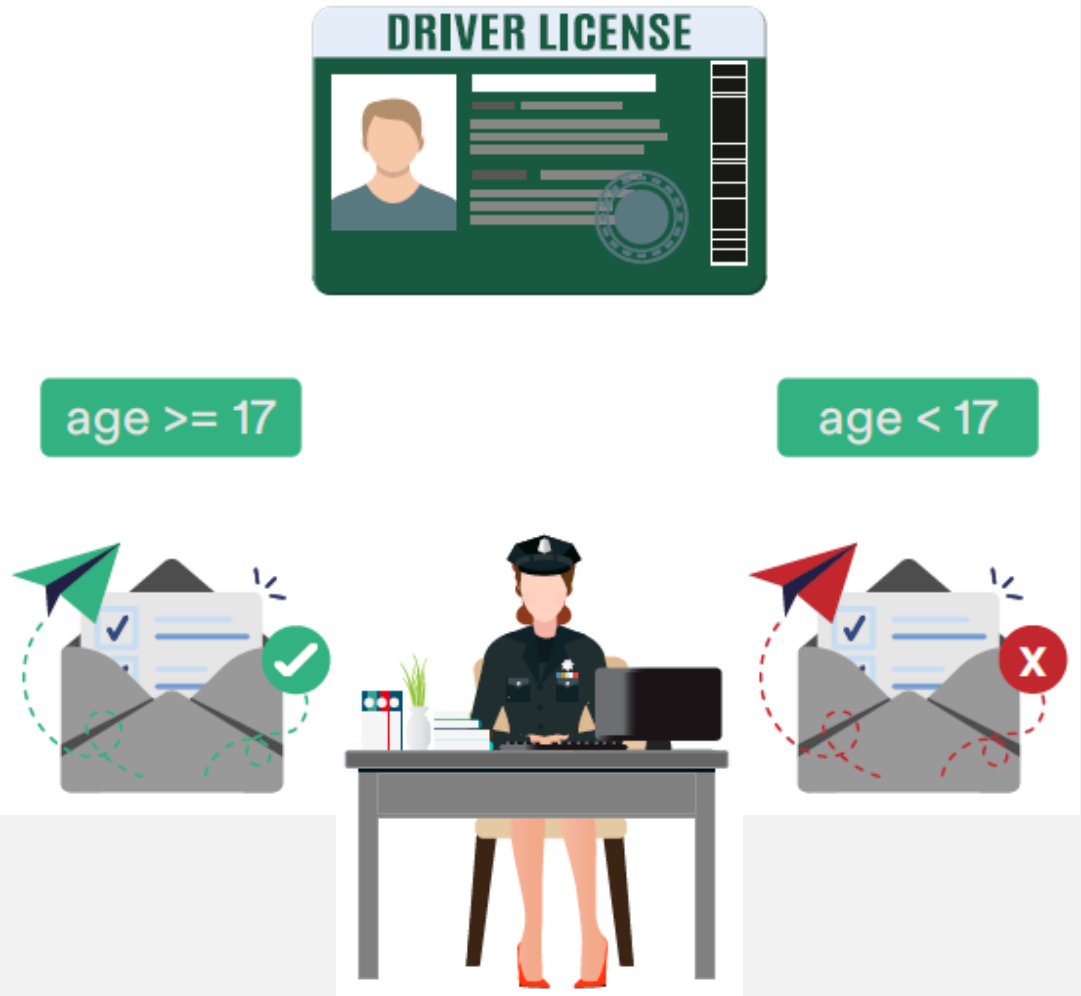
Statements that will be executed when condition is True

indentation

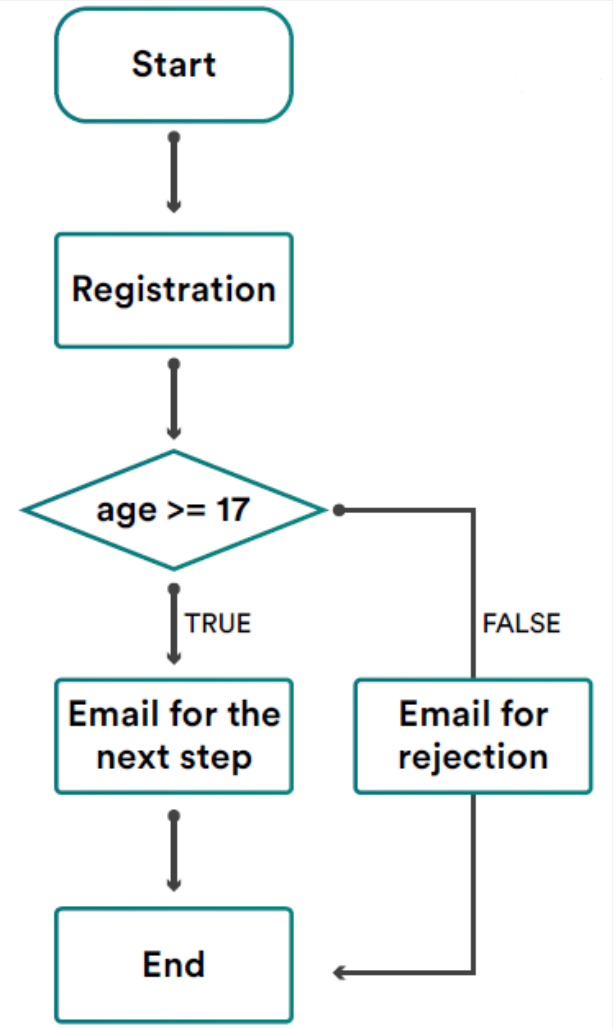


if else Statement

if else Statement



if else Statement

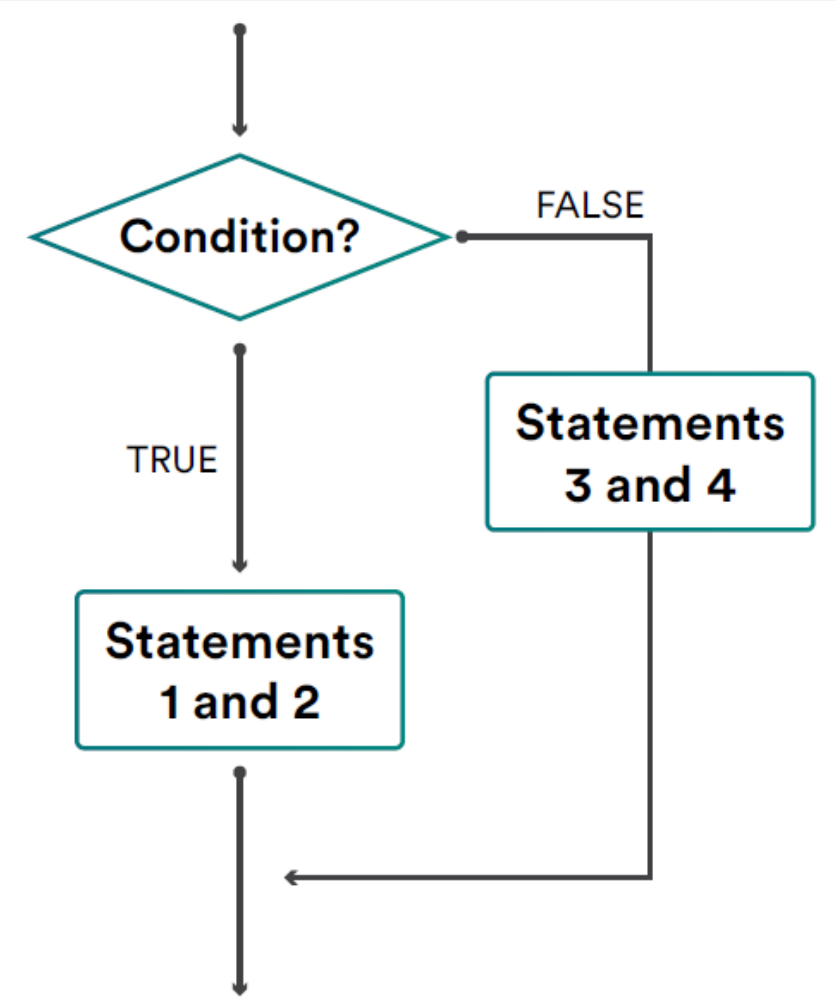


if else Statement

SYNTAX

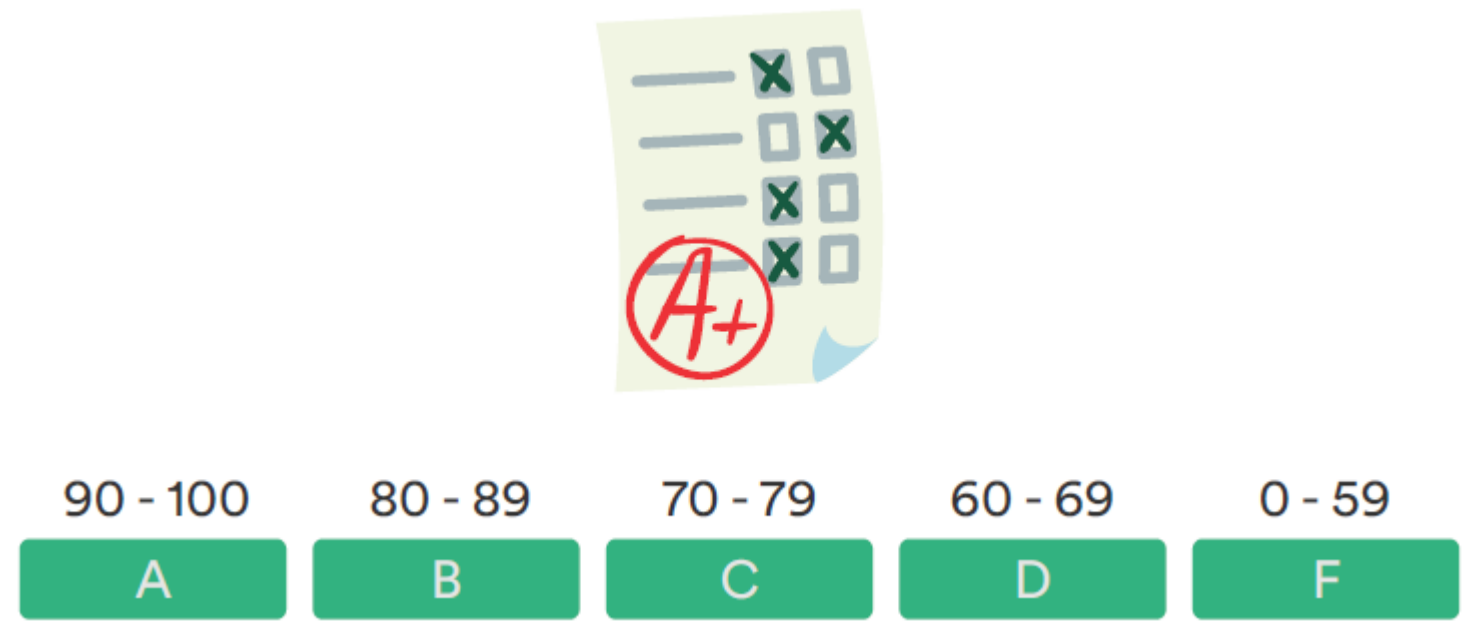
```
if condition :
    statement 1
    statement 2
else :
    statement 3
    statement 4
```

indentation

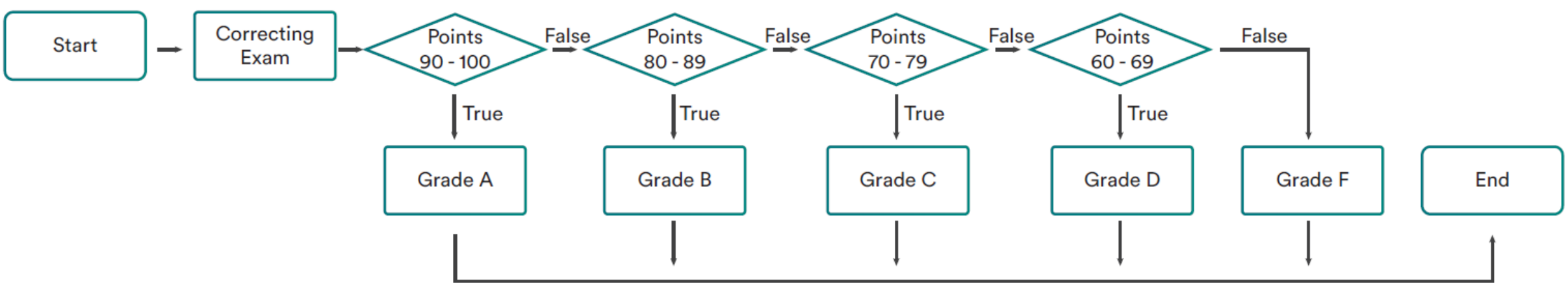


if elif else Statement

if elif else Statement



if elif else Statement

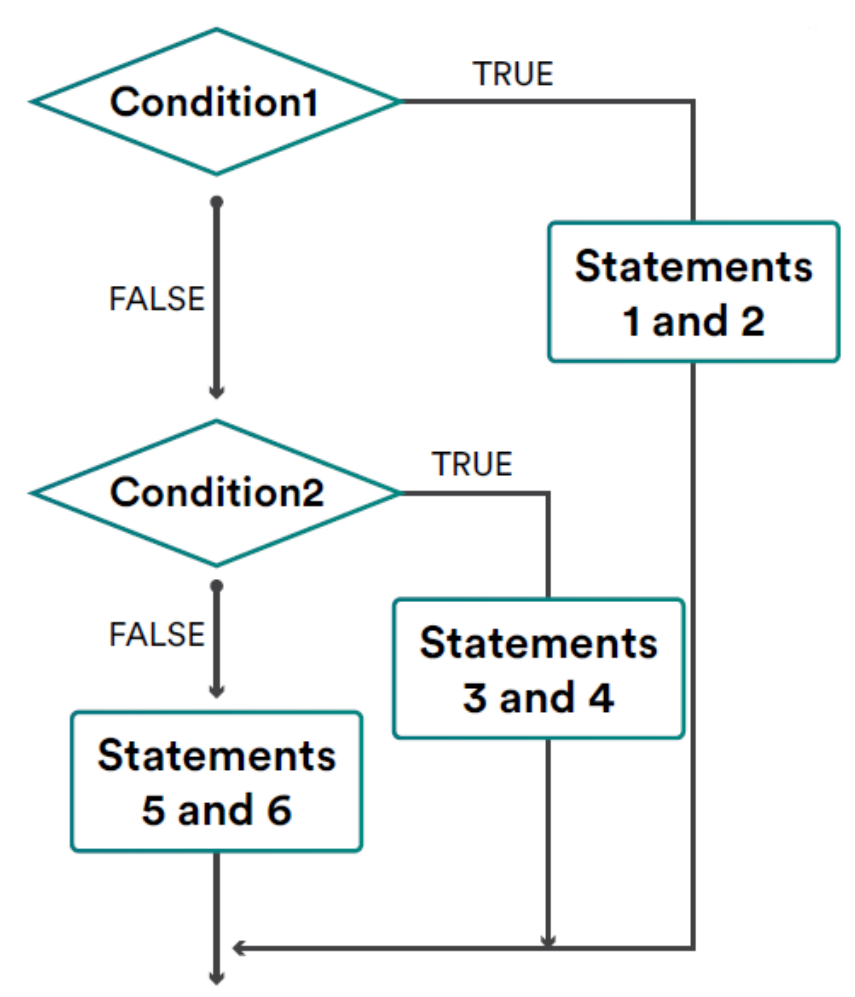


if elif else Statement

SYNTAX

```
if condition :
    statement 1
    statement 2
elif condition:
    statement 3
    statement 4
else :
    statement 5
    statement 6
```

indentation



Exercise

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Ergonomie mobile
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Soal 1

if $x = 4$, $y = 3$ & $z = 2$

$$w = \left(\frac{x + y \times z}{x \times y}\right)^{z} = ?$$

Soal 2

```
Silahkan masukkan angka berapapun :  5
Kuadrat dari 5 = 25
```

Soal 3

485 hari.

Nyatakan dalam tahun, bulan, minggu dan hari.

*1 bulan = 30 hari, 1 tahun = 360 hari.

Soal 4

Saat ini, jumlah usia Andi & Budi = 49 th,

dengan rasio Usia Andi & Budi = 0.4.

Berapa usia Andi & Budi 2 tahun lagi?

Soal 5

Jarak mobil A & B = 120 km.

A berjalan 60km/h dari timur.

B berjalan 40km/h dari barat.

A & B start pukul 9 WIB.

Jam berapa A & B bertabrakan?

Soal 6

Masukkan Jumlah Apel : 2
Masukkan Jumlah Jeruk : 3
Masukkan Jumlah Anggur : 5

Detail Belanja

Apel : 2 × 10000 = 20000
Jeruk : 3 × 15000 = 45000
Anggur : 5 × 20000 = 100000

Total : 165000

Soal 7

```
PS D: \Purwadhika\Purwadhika\Python Fundamental> python fundamental.py
Masukkan angka : 7
Angka 7 tergolong bilangan GANJIL!
PS D: \Purwadhika\Purwadhika\Python Fundamental> python fundamental.py
Masukkan angka : 6
Angka 6 tergolong bilangan GENAP!
```

Soal 8

IMT = massa(kg) / tinggi(meter)^2
a. IMT < 18.5 artinya berat badan kurang,
b. 18.5 - 24.9 artinya berat badan ideal,
c. 25.0 - 29.9 artinya berat badan berlebih,
d. 30.0 - 39.9 artinya berat badan sangat berlebih,
e. IMT > 39.9 artinya obesitas.

Masukkan Massa (kg) : 67

Masukkan Tinggi (cm) : 178

Massa 67 kg dan tinggi 1.78 m

IMT = 21.1463199909102385, BERAT BADAN IDEAL!

Soal 9

Condition 1

Masukkan Jumlah Apel : 3
Masukkan Jumlah Jeruk : 2
Masukkan Jumlah Anggur : 5

Detail Belanja
Apel : 3 × 10000 = 30000
Jeruk : 2 × 15000 = 30000
Anggur : 5 × 20000 = 100000

Total : 160000

Masukkan jumlah uang : 135000
Transaksi anda dibatalkan
uangnya kurang sebesar 25000

Condition 2

Masukkan Jumlah Apel : 3
Masukkan Jumlah Jeruk : 2
Masukkan Jumlah Anggur : 5

Detail Belanja
Apel : 3 × 10000 = 30000
Jeruk : 2 × 15000 = 30000
Anggur : 5 × 20000 = 100000

Total : 160000

Masukkan jumlah uang : 160000
Terima kasih

Condition 3

Masukkan Jumlah Apel : 3
Masukkan Jumlah Jeruk : 2
Masukkan Jumlah Anggur : 5

Detail Belanja
Apel : 3 × 10000 = 30000
Jeruk : 2 × 15000 = 30000
Anggur : 5 × 20000 = 100000

Total : 160000

Masukkan jumlah uang : 175000
Terima kasih

Uang kembali anda : 15000

Thank You!

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